

Unit 3, Lesson 8: Analyzing Residuals



Benchmark

MA.912.DP.2.6 Given a scatter plot with a line of fit and residuals, determine the strength and direction of the correlation. Interpret strength and direction within a real-world context.



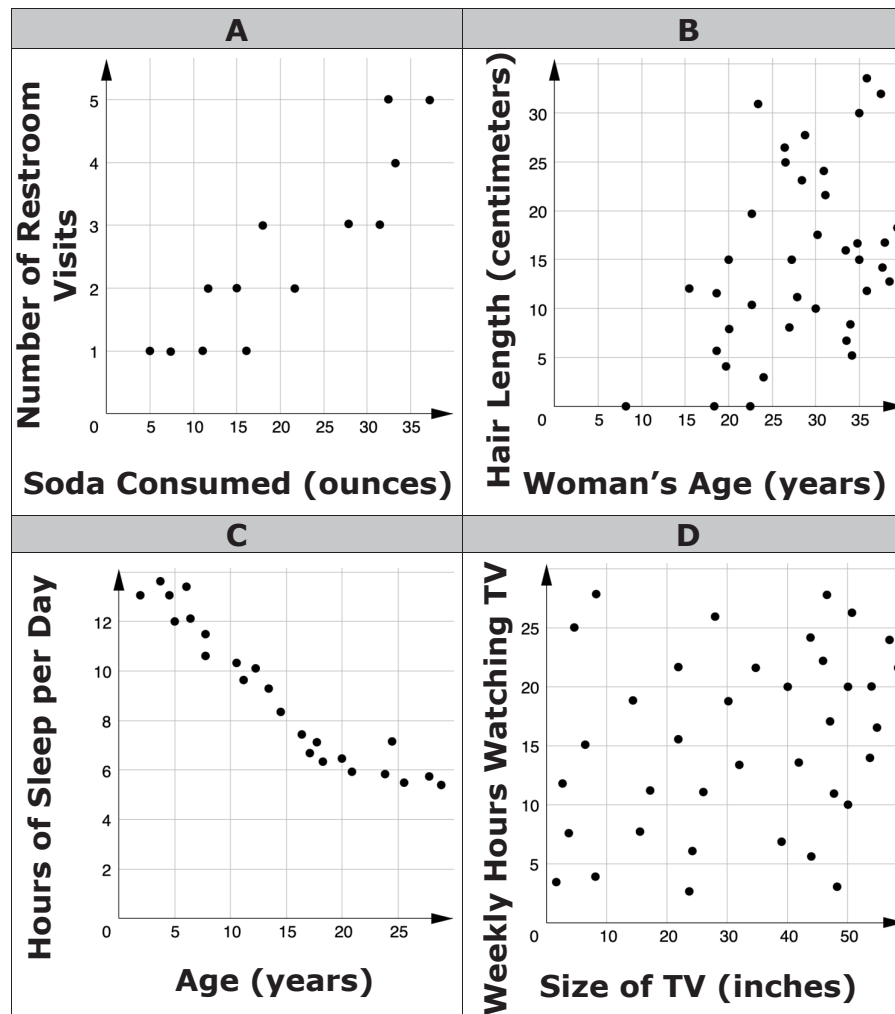
Learning Targets

- ☐ I can determine whether a function is an appropriate fit for the data by analyzing residuals.
- ☐ I can interpret the strength and direction of the correlation within a real-world context.



3.8.1 Warm-Up: Bell Work

Four scatter plots are shown.



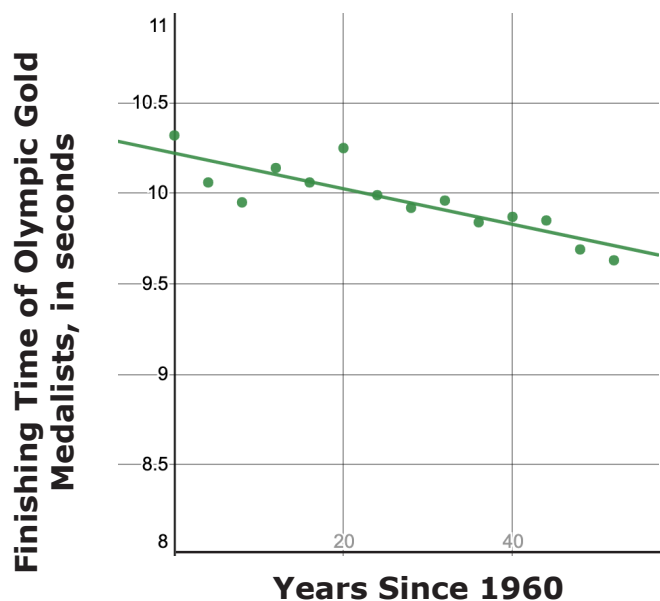
1. Order the scatter plots from bivariate data that is "least closely fit to a line" to "most closely fit to a line" by writing the letter corresponding to each graph.

Least Closely Fit to a Line		Most Closely Fit to a Line	



3.8.2 Exploration: Predicted vs. Actual Values

The scatter plot below shows the finishing times for the Olympic gold medalist in the men's 100-meter (m) dash for the Olympic games since 1960.



The raw data used to construct the scatter plot is shown.

Year of Olympics	1960	1964	1968	1972	1976	1980	1984
Gold Medal Time (in seconds)	10.32	10.06	9.95	10.14	10.06	10.25	9.99
Year of Olympics	1988	1992	1996	2000	2004	2008	2012
Gold Medal Time (in seconds)	9.92	9.96	9.84	9.87	9.85	9.69	9.63

The equation of the line of best fit that models the relationship is $f(x) = 10.2217 - 0.0098x$, where f is the finishing time of the Olympic gold medalist in the men's 100-meter dash and x is the number of years since 1960. Work with your partner to complete the following using the information on the previous page.

- Consider the 2012 Summer Olympic games.
 - How many years after 1960 was the 2012 Summer Olympics?
 - Using the equation that models the relationship between years since 1960 and the Olympic gold medal finishing times, what is the predicted finishing time for the men's gold medalist in 2012, rounded to the nearest hundredth?
 - What was the actual finishing time for the men's gold medalist in 2012?
 - Complete the statement using your answers from Parts B and C.

The actual finishing time for the men's gold medalist in the 2012 Summer Olympics was

☐ faster
☐ slower than predicted using the line of best fit
by _____ seconds.

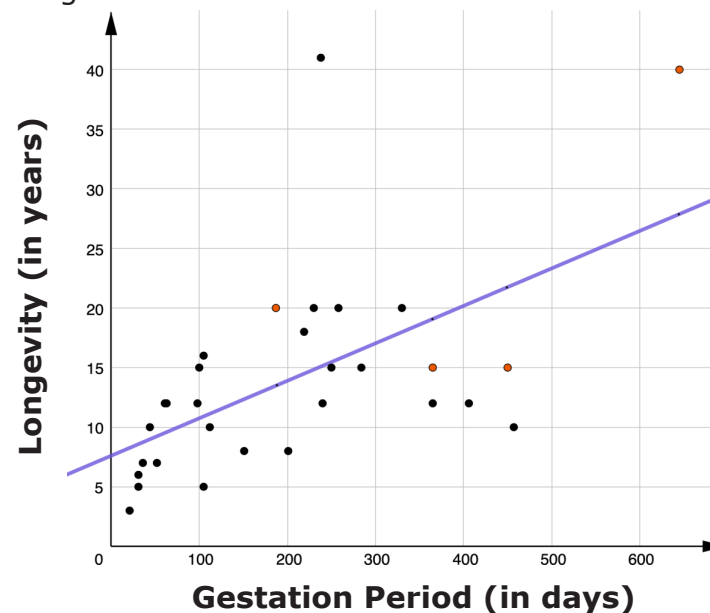
2. Identify one point on the scatter plot where the actual finishing time for the gold medalist was more than the model predicted.
 - a. Find the coordinates of the point.
 - b. Explain how you know the actual finishing time was more than the model predicted.

3. Discuss with your partner whether the linear function appears to be a good fit for the data and how you know. Summarize your discussion.



3.8.3 Guided Instruction: Analyzing Residuals

The scatter plot summarizes the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives.



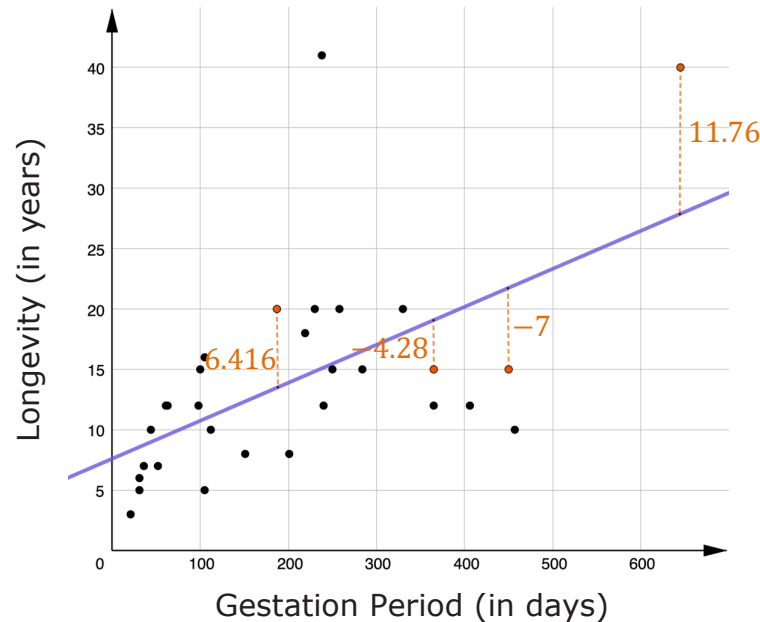
The function $g(x) = 0.032x + 7.6$ is fit to the data.

1. Consider the table shown with the x -values, actual y -values, predicted y -values, and residual values of the **orange points** on the scatter plot.

x -value	Actual y -value	Predicted y -value	Residual Value
187	20	13.584	6.416
365	15	19.28	-4.28
450	15	22	-7
645	40	28.24	11.76

- a. Discuss with your partner what you think a residual value is.

The scatter plot is shown, with the residual values of the four points from the table labeled.



- b. Complete the statements.

A residual value is the difference,

- ☐ (actual y -value) - (predicted y -value)
☐ (actual y -value) + (predicted y -value) for a

given x -value in the data set. A residual value

- ☐ close to
☐ far from 0 indicates the actual data value is

close to the line of best fit.

A positive residual value means the actual point lies

- ☐ on
☐ above
☐ below

the line of best fit. A negative residual

value means the actual point lies

- ☐ on
☐ above
☐ below

the line

of best fit. A residual value of 0 means the actual

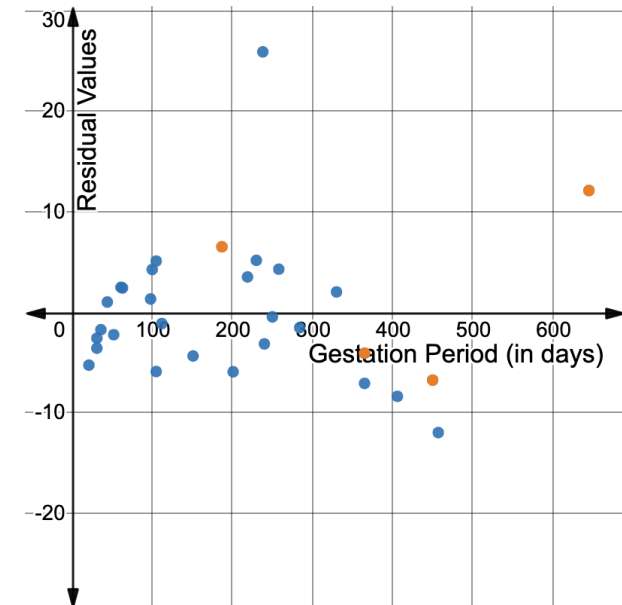
point lies

- ☐ on
☐ above
☐ below

the line of best fit.

Residuals are used to analyze the strength of a line's fit to the data. They can be represented on a residual plot like the one shown below.

Each point on the residual plot shown shows the residual value for a corresponding point on the scatter plot. The residual values explored in Problem 1 are shown in orange.

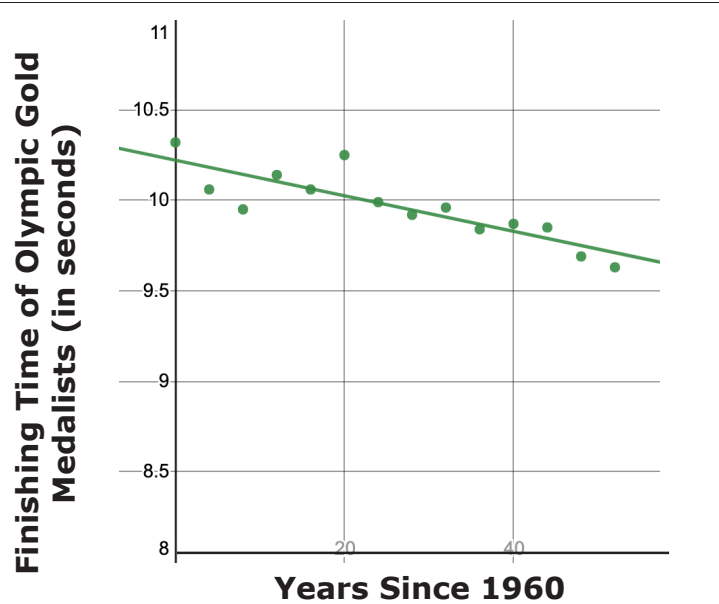




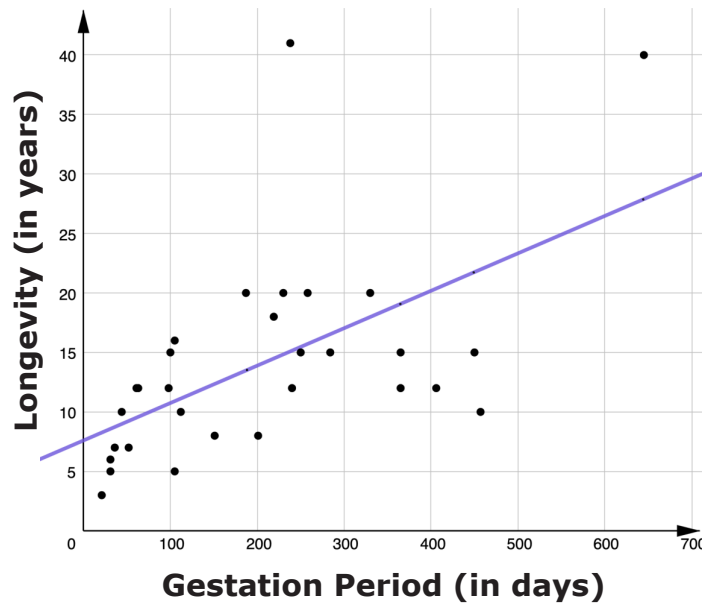
3.8.4 Bring It Together: Interpreting the Strength and Direction of a Line of Fit

The two scatter plots explored in this lesson are shown.

Scatter Plot A

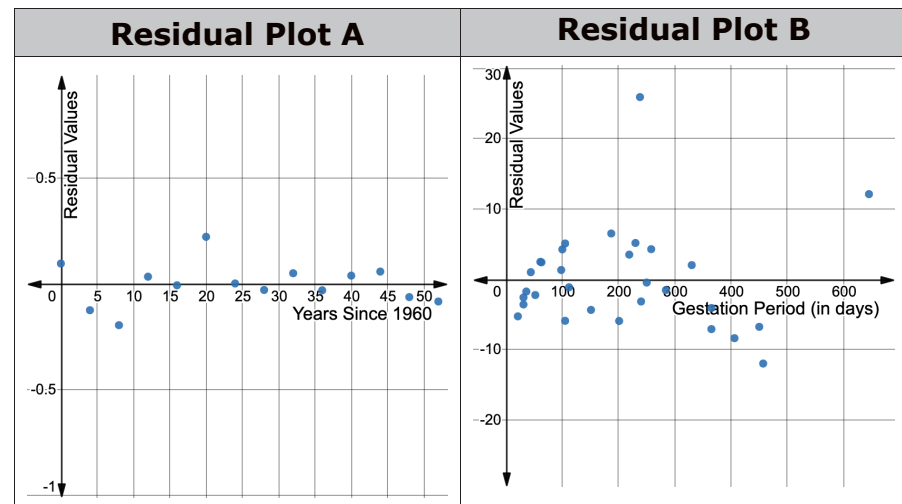


Scatter Plot B



1. Explain which data appears to have a stronger linear association: the data represented on scatter plot A or scatter plot B?

2. The residual plots for both scatter plots are shown.



Explain how the residual values shown on the residual plots above support your answer to Problem 1.

3. The linear functions fit to each set of data are shown.

Linear Function Fit to Scatter Plot A	Linear Function Fit to Scatter Plot B
$f(x) = 10.2217 - 0.0098x$	$g(x) = 0.032x + 7.6$

Complete the statements.

The slope of the function fit to scatter plot A is

- ☐ negative,
- ☐ positive,

which means the finishing time for

Olympic gold medalists in the men's 100-m dash

generally

- ☐ increases
- ☐ decreases

as the number of years since

1960 increases. The strength of this linear relationship

is

- ☐ strong
- ☐ weak

because the residual values are

- ☐ relatively small
- ☐ relatively large

and

- ☐ randomly scattered around the x -axis.
- ☐ show a pattern.

The slope of the function fit to scatter plot B is

- ☐ negative,
- ☐ positive,

which means the longevity of an

animal's life generally

- ☐ increases
- ☐ decreases

as the gestation

period increases.

The strength of this linear relationship is

- ☐ strong
- ☐ weak

because the residual values are

- ☐ relatively small
- ☐ relatively large

and

- ☐ randomly scattered around the x -axis.
- ☐ show a pattern.



3.8.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate how you feel about this lesson's learning targets.

- ☐ I can determine whether a function is an appropriate fit for the data by analyzing residuals.
- ☐ I can interpret the strength and direction of the correlation within a real-world context.

